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RV COLLEGE OF ENGINEERING®

Autonomous Institution affiliated to VTU

III Semester B. E. April 2024 Examinations

DEPARTMENT OF INDUSTRIAL ENGINEERING AND MANAGEMENT**WORK SYSTEMS DESIGN****(2022 SCHEME)****MODEL QUESTION PAPER****Time: 03 Hours****Maximum Marks: 100****Instructions to candidates:**

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2 is compulsory. Answer any one full question from 3 and 4, 5 and 6, 7 and 8 and 9 and 10.

PART – A (Objective type for one or two marks)**(True & false and match the following questions are not permitted)**

1	1.1	Distinguish between the terms: Production and Productivity	02
	1.2	Classify the types of Worker-Machine Systems	02
	1.3	Categorize the features and specifications to be considered in selecting among alternative improvement proposals in Methods Engineering.	02
	1.4	Expand the acronym USA in the context of methods engineering and automation.	02
	1.5	What is meant by the term Application Speed Ratio?	02
	1.6	Mention the prerequisites of Standard Time.	02
	1.7	What is meant by Lean Production System?	02
	1.8	Distinguish between Pull System and Push System of Production Control.	02
	1.9	List the three categories of Human-Machine Systems.	02
	1.10	What is meant by Anthropometry?	02

PART – B (Maximum subdivision is limited to 4 in each question)

UNIT – I			
2	a	Distinguish between basic productive work environment and Excess non productive work environment. Also classify the excess non productive activities into three categories. Support your explanation with examples for each category.	08
	b	The normal time for a repetitive task that produces two work units per cycle is 3.0 minutes. The plant uses a PFD allowance factor of 15%. Determine (a) the standard time per piece and (b) how many work units are produced in an 8 hour shift at standard performance.	08

UNIT - II			
3	a	Review any two categories of specialized analysis techniques used in Methods Engineering that are exclusive to operations analysis and industrial engineering.	08
	b	Identify and explain the types of symbols used in a flow process chart.	08
OR			
4	a	Describe any two basic data collection and analysis techniques used in Methods Engineering.	08

	b	Enumerate the principles of motion economy related to the design of tooling and equipment.	08
UNIT - III			
5	a	Discuss the characteristics typical of industrial situations in which time standards would be beneficial?	08
	b	The standard time is to be established for a manual work cycle by direct time study. The observed time for the cycle averaged 4.80 minutes. The worker's performance was rated at 90% on all cycles observed. After eight cycles, the worker must exchange parts containers, which took 1.60 minutes, rated at 120%.The PFD allowance for this class of work is 15%.Determine (a) normal time and standard time (b) worker's efficiency if the worker produces 123 work units during an 8 hour shift.	08
OR			
6	a	Summarize different varieties of Predetermined Motion Time Systems.	08
	b	Review different versions of MOST work measurement packages.	08

UNIT - IV			
7	a	Provide a historical note on the Lean Production Philosophy.	08
	b	With the help of an illustration, explain the working of a kanban system between work stations.	08
OR			
8	a	Explain any two aspects of Autonomation with suitable examples.	08
	b	What is Standardized Work Procedure? Explain the components of the same for a given task.	08

UNIT - V			
9	a	What is human machine system? Review the two main components of the same.	08
	b	Discuss the components of Physical work environment.	08
OR			
10	a	What is an industrial accident? Analyze the causes for the same from ergonomics perspective.	08
	b	Describe any four common occupational diseases along with necessary causes.	08